Deep Demonstration Tracing: Learning Generalizable Imitator Policy for Runtime Imitation from a Single Demonstration

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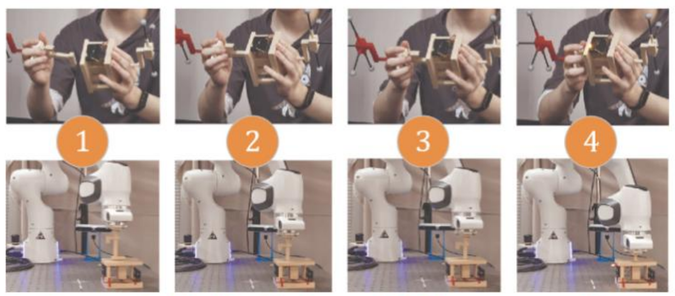
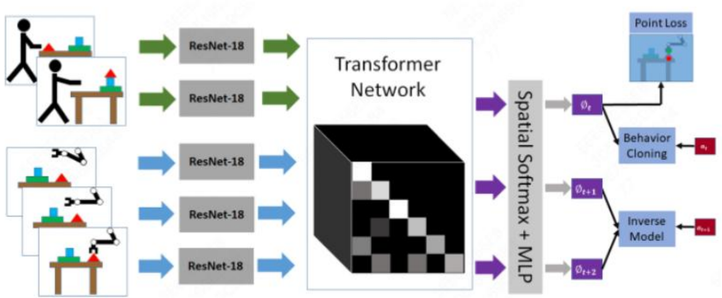
Introduction

The vision of Runtime One-shot imitation Learning

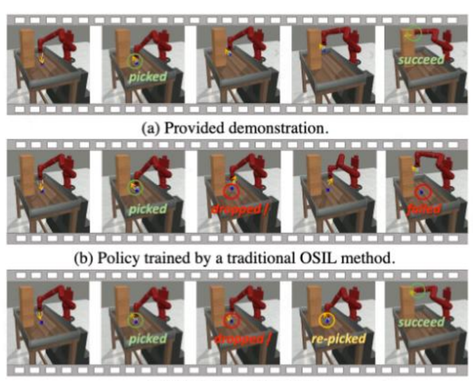


Achieve any tasks directly "prompted" by corresponding demonstration.

Popular paradigm: Transformer + behavior cloning



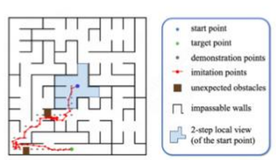
Our focus: Generlization Challenges of runtime OSIL



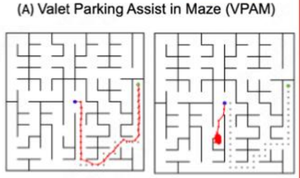
(a) Provided demonstration.

(b) Policy trained by a traditional OSIL method.

(c) Policy trained by DDT.



(A) Valet Parking Assist in Maze (VPAM)



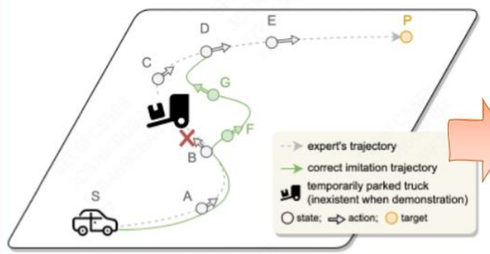
Trained by DDT

Trained by traditional OSIL

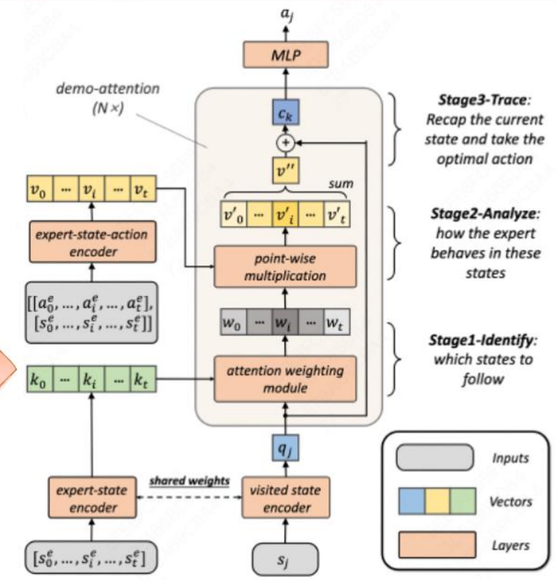
- ***Unseen demonstrations in unseen envirnoments.***
  -> incorrect representation caused by transformer.
- ***Unforseen changes after demonstrations collection.***
  -> limitation of behavior cloning.

Methodology

KEY1: Inject the inductive bias of "how human make decisions in runtime OSIL" into the imitator policy network.

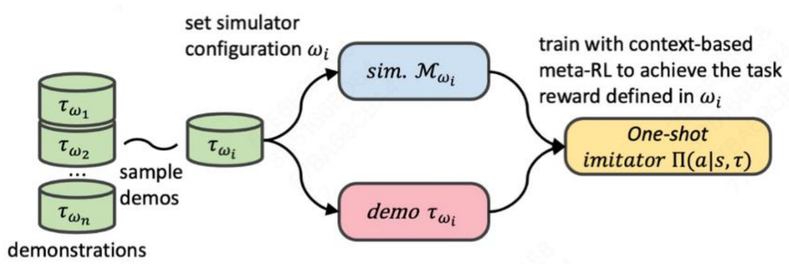


An example of 3-stage OSIL of humans.



- Stage 1: **Identify** relevant states within the trajectory based on the current state.
- Stage 2: **Analyze** the expert's behavior patterns associated with these states.
- Stage 3: **Trace** the expert's demonstrations based on the relationship between the current state and the expert's behavior patterns in the demonstrations.

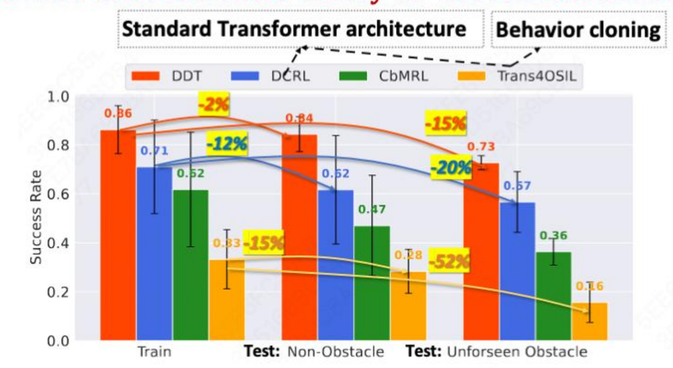
KEY2: Solve runtime one-shot imitation learning by context-based meta-RL, instead of supervised learning



- The unforeseen changes will randomly apprear in the simulators ($\mathcal{M}$).
- With meta-RL, the imitator policy will try to achieve *all of* the targets the same to the demonstration guided by 0-1 task rewards.
- In the process, the imitator policy will suffer from the unforseen changes and *have to handle them before achieve the targets.*

Major Experiments

Q1: One-shot imitation ability in unseen situations



Group results averged by 8 settings with 3 seeds (VPAM env).

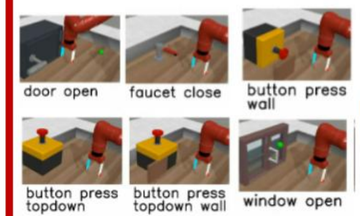
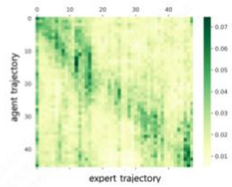


Table 4: Performance on unseen heterogeneous demonstrations.

Environment	Button Press	Door Close	Reach
Performance	0.78	1.00	0.75

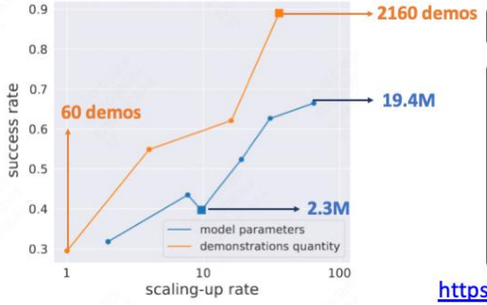
Q2: Does DDT imitating via tracing the demos

(a) Trajectory of DDT.



(b) Attention score.

Q3: The "Scaling Law" of DDT in the OSIL setting



MORE VIDEO

https://osil-ddt.github.io/